UNRAMIFIEDNESS OR LOCAL QUASI-FINITENESS OF A COVER FORCES THE SAME PROPERTY FOR THE DIAGONAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The statement that an unramified flat morphism of finite presentation of algebraic spaces is etale. Invoked at the start of the counted proof (source lines 1129-1131) as a cross-chapter pointer. Explains the parenthetical clause of the statement.
- The criterion for checking a property of a morphism of algebraic stacks along a smooth surjective cover. Invoked inside the counted proof (source lines 1140-1145) as a cross-chapter pointer. Reduces the property of the diagonal to the corresponding property of the comparison morphism.
- The identification of the pullback of the diagonal with the fibre product of the covering algebraic space with itself. Used inside the counted proof (source lines 1137-1139). Turns the comparison morphism into the canonical map from the fibre product to the square.
- The permanence statements for unramified and for locally quasi-finite morphisms of algebraic spaces. Invoked at the end of the counted proof (source lines 1146-1155) as cross-chapter pointers. Gives unramifiedness or local quasi-finiteness of the canonical map from the two projections.

Source reference. Original reference spaces-morphisms-lemma-permanence-quasi-finite: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 971–971. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-permanence-quasi-finite, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference spaces-morphisms-lemma-permanence-unramified: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 968–968. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-permanence-unramified, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-unramified-flat-lfp-etale: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1131–1131. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-unramified-flat-lfp-etale, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-lemma-check-property-covering: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 479–479. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-check-property-covering, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. Let W be an algebraic space, and let $f : W \to \mathcal{X}$ be a surjective, flat, locally finitely presented morphism.*

(1) *If f is unramified (i.e., étale, i.e., $\mathcal{X}$ is Deligne-Mumford), then $\mathcal{X}$ is DM.*
(2) *If f is locally quasi-finite, then $\mathcal{X}$ is quasi-DM.*

Proof. Note that if f is unramified, then it is étale by Morphisms of Spaces, Lemma spaces-morphisms-lemma-unramified-flat-lfp-etale. This explains the parenthetical remark in (1). Assume f is unramified (resp. locally quasi-finite). We have to show that $\Delta_{\mathcal{X}} : \mathcal{X} \to \mathcal{X} \times \mathcal{X}$ is unramified (resp. locally quasi-finite). Note that $W \times W \to \mathcal{X} \times \mathcal{X}$ is also surjective, flat, and locally of finite presentation. Hence it suffices to show that

$$W \times_{\mathcal{X} \times \mathcal{X}, \Delta_{\mathcal{X}}} \mathcal{X} = W \times_{\mathcal{X}} W \longrightarrow W \times W$$

is unramified (resp. locally quasi-finite), see Properties of Stacks, Lemma stacks-properties-lemma-check-property-covering. By assumption the morphism $\mathrm{pr}_i : W \times_{\mathcal{X}} W \to W$ is unramified (resp. locally quasi-finite). Hence the displayed arrow is unramified (resp. locally quasi-finite) by Morphisms of Spaces, Lemma spaces-morphisms-lemma-permanence-unramified (resp. Morphisms of Spaces, Lemma spaces-morphisms-lemma-permanence-quasi-finite). □